Heimdall: Real-Time, Bank-Side Holds on UPI Mule Money

AIMIT FinTech Innovation Hackathon 2026 · Track 1: Digital Payments Fraud Detection & Prevention

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Code: `github.com/gokeeel/heimdall-upi` · Notebook: `kaggle.com/code/gokuljayachandran/heimdall-benchmark`

Problem statement. UPI carried 24.07 billion transactions worth Rs 29.37 lakh crore in September 2026 [1]. Fraudsters route stolen money through *mule accounts*, which receive a victim's money and pass it on within hours; banks usually act only after a complaint, when the money has gone. RBI's draft KYC Amendment Directions (11 Sep 2026; banks to implement by 1 Apr 2027) let banks temporarily **hold suspected mule transactions of Rs 1,000 and above**, with immediate notice, 20 days for the customer to explain, a decision within 10 days of the reply (or 30 days) and a 60-day cap [2]. A hold prevents loss only if the bank decides *during the payment*, which NPCI caps at 15 seconds [3].

Proposed solution. **Heimdall** sits inside the *receiving* bank, next to its existing fraud checks; nothing changes at NPCI. For every account it keeps a small **risk record**: running numbers such as how many different contacts it deals with, the share of contacts where money flows both ways, and money sent out versus received, updated by every transaction. A nightly batch job adds **link numbers**: how many accounts share its counterparties or address (two hops) and its links to accounts the bank already flagged (an internal **suspect list**). When a credit arrives, a model trained in advance turns the record into a **risk score** and the bank takes one of three actions (Figure 1): **real time means the action happens during the payment**.

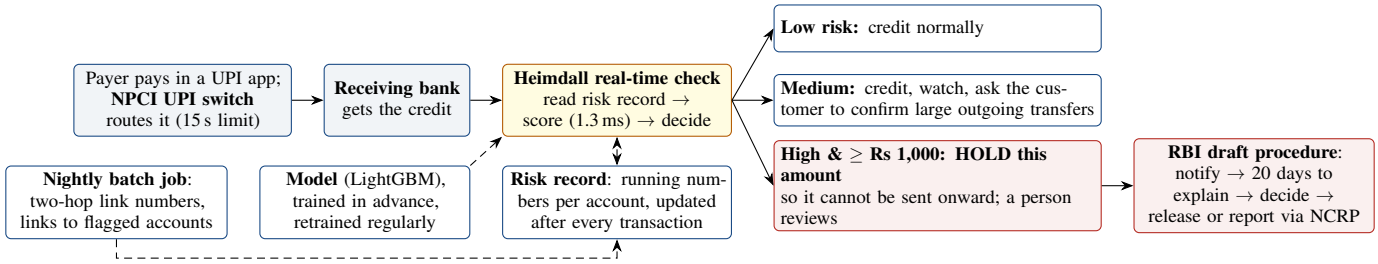


Figure 1. Where Heimdall sits in one UPI payment, and the three actions.

Innovation. (1) **Prevention at the first hop:** hold the amount as it lands in a likely mule account, before it is forwarded. (2) **An honest test, measured in money:** models learn only from 2022–2024 and are tested once on 2025, and we report the share of mule *money* held, not just accounts flagged; public NFPC solutions we examined use full histories and random splits, which flatter results. (3) **Point-in-time links:** every link number uses only data and flags from before the scoring day. (4) **Safeguards:** hold levels set by review capacity, a person on every hold, fairness and planted-trap checks.

Feasibility. Each bank checks only its own customers, so UPI's average load of about 9,300 transactions per second is shared. Scoring one account takes **1.3 ms** on a 2-core cloud CPU, a tiny part of the 15-second window; updating the record takes a few additions, and link numbers are precomputed nightly, so no graph model runs in the payment. The full per-transaction engine and its memory cost (e.g. Redis for crores of accounts) are the next stage. Precedents: Brazil's Pix lets the receiving institution block suspected-fraud funds for 72 hours [4], UK banks may delay a payment by up to 4 business days [5], and Mastercard scores UK bank-to-bank payments against mule networks in real time [6]. Of four methods tested on the RBIH NFPC dataset [7] (Table 1), we chose LightGBM: it holds as much mule money as any at equal cost and is 45× faster than a random forest.

Method (same 28 features, default settings)	ROC-AUC	PR-AUC	Mules caught at 1 false alarm per 100 genuine	… per 1,000	Mule money held, at 1 in 100 genuine customers held	Time per account
Simple rule (no learning)	0.81	0.08	3%	0.2%	3%	–
Logistic regression	0.81	0.15	14%	4%	11%	1.3 ms
Random forest	0.81	0.17	19%	6%	15%	56 ms
LightGBM (chosen)	**0.83**	**0.14**	**14%**	**4%**	**17%**	**1.3 ms**
Check: LightGBM, own behaviour only (no links)	0.83	0.15	14%	2%	13%	0.9 ms
Check: same LightGBM, usual random 70/30 split	0.91	0.55	64%	19%	–	–

Table 1. Learn from 2022–2024 snapshots (239,816 account-snapshots; 319 mules); test once on 1 Jan 2025 (92,877 accounts; 1,767 flagged during 2025). ROC-AUC: how often a mule outscores a genuine account (0.5 = coin toss; LightGBM 95% range 0.81–0.84). PR-AUC: share of alerts that are mules (coin toss ≈ 0.02). Money: Jan–Jun 2025 replay, re-scored monthly; share of credits ≥ Rs 1,000 reaching mules before the bank's own flag (LightGBM range 13–22%).

Impact. In the Jan–Jun 2025 replay, with the hold level fixed beforehand on pre-2025 data, Heimdall would have **held 23% (18–28%) of the mule money** that arrived before the bank's own flag (Rs 968 crore, synthetic), with holds on 310 of 1,767 mules a median of **151 days before** the bank flagged them. The cost: 1,514 genuine customers (1.7%, above the 1 in 100 aimed for) would have had a hold for a person to review, so the level should be re-set monthly by review capacity. **Honest testing matters:** a random split of the same accounts reports PR-AUC 0.55, not 0.14, as the model then learns from the same 2025 sweep. The public DhanRakshak solution reports AUC 0.967 on the competition leaderboard and 0.920 in its own held-out run, using full account histories, including months of mule activity [8]; Heimdall answers an earlier, harder question. **Links** gave no consistent gain here: own behaviour alone ties on 2025 and is slightly better on validation in 2022–2024, while links do better at 1 in 1,000 and on money held (17% vs 13%, overlapping ranges). PIN-code/branch shares lowered ROC-AUC to 0.78 and could penalise neighbourhoods, so are excluded. *Limits:* synthetic data; labels are the bank's flags, mostly from one Apr–Jun 2025 sweep, with only 319 flagged before 2025; the replay re-scores monthly, not live; genuine rural customers were wrongly flagged 1.2× as often as urban (0.66% vs 0.54%), so rural flags go to a person first; real-world validation needs a partner bank.

Technology. Python, pandas, scikit-learn, LightGBM; a public Kaggle notebook rebuilds every number from all 396 data files in about 30 minutes; code on GitHub (MIT licence). Production path: hook behind the bank's UPI switch → stream processing (e.g. Kafka) → in-memory risk records (e.g. Redis) fed by a nightly link job → LightGBM scorer → action rules → officer case screen and NCRP referral, with a plain-language reason for each hold. Data: RBIH & IIT Delhi NFPC Phase-2 (CC BY-NC-SA 4.0); no confidential or real customer data.

Target beneficiaries. UPI users and fraud victims, whose money is stopped before it leaves; genuine customers, who face a held amount with a reason and deadline, not a frozen account; bank fraud and compliance teams, who get an RBI-SOP-ready workflow at a controlled workload; law enforcement, via NCRP referrals; and the ecosystem: the suspect-list slot can also take external lists like the I4C suspect registry or MuleHunter.AI.

Declaration: this submission is our own original work; all data and prior work used are cited.

References. [1] NPCI UPI statistics, Sep 2026. [2] RBI, Draft KYC (Amendment) Directions 2026, debit-hold SOP, 11 Sep 2026. [3] NPCI, UPI-OC No. 214, 26 Apr 2025. [4] Banco Central do Brasil, 72-hour Pix fraud block (InfoMoney, 29 Sep 2021). [5] UK Payment Services (Amendment) Regulations 2024. [6] Mastercard press release, 6 Jul 2023. [7] RBIH & IIT Delhi, NFPC Phase-2 dataset, Kaggle `abhyudayrbih/rbih-nfpc-phase-2`. [8] S. Singh, DhanRakshak, `github.com/shivangiS04/DhanRakshak` (README, `verified_metrics.json`).